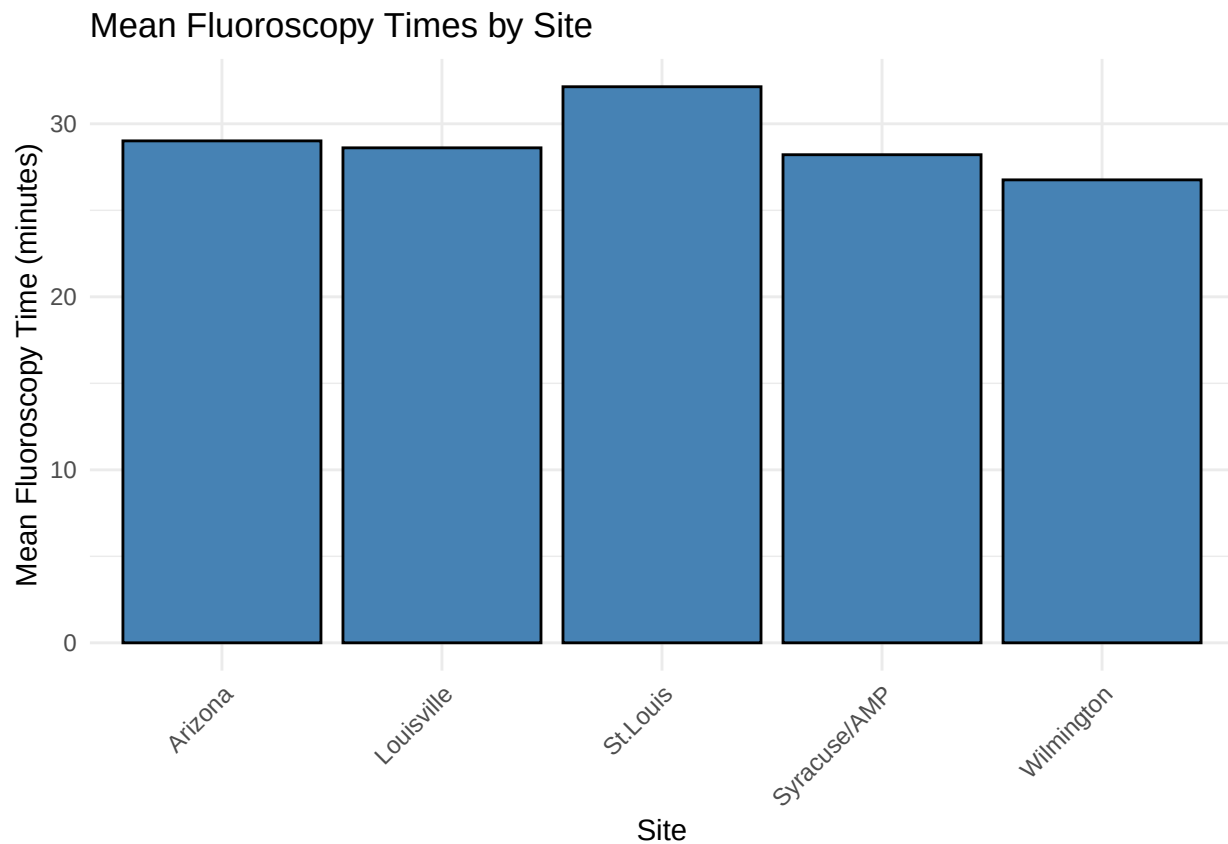


IR Center Project

Nicholas Kaufman

2025-05-14

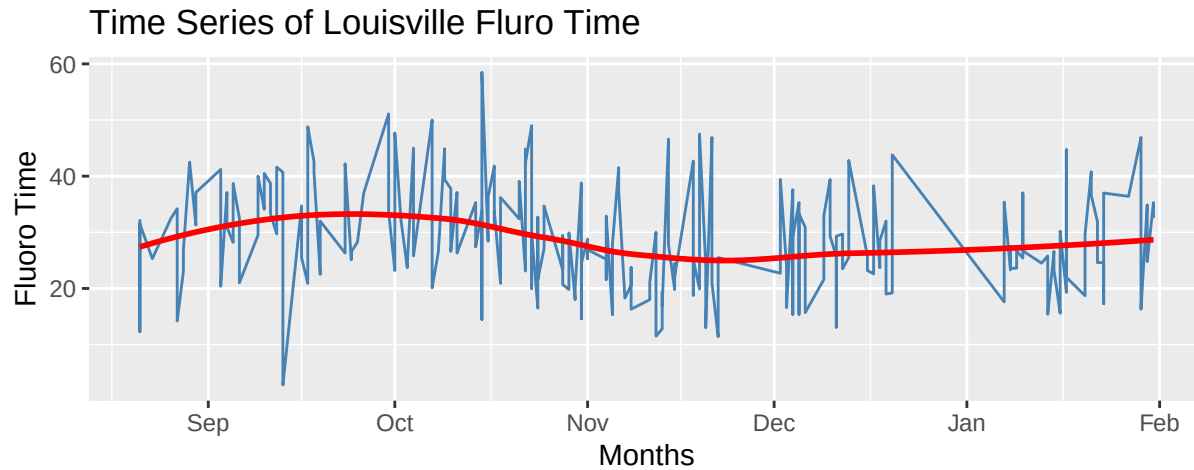


If we're looking at the data from the most broad viewpoint, we see that Wilmington had the lowest Fluoro times averaged out over the study duration, with St. Louis, having the highest. Though it has yet to be determined whether these differences are significant. Despite this, it would be much more beneficial to look at the data as a progression rather than a blanket average as a whole.

Let's look at each location individually as a time series progression.

Louisville

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Month	Mean_Fluoroscopy_Time
August 2024	28.60000
September 2024	32.90588
October 2024	30.99608
November 2024	25.27727
December 2024	27.01707
January 2025	27.28810

Looking at the time series, it's difficult to infer any real difference in fluoro time over the year. We'd expect a provider's performance to increase (inversely decreasing procedure time) over the course of the year, but there seems to be small, most likely arbitrary differences in mean fluoroscopy time at the Louisville site. However, it is worth a look to see if any of the differences are statistically significant.

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Month      5   1584    316.8   3.877 0.00219 **
## Residuals 218  17813     81.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Fluoroscopy_Time ~ Month, data = Louisville)
##
## $Month
##           diff          lwr          upr      p adj
## September 2024-August 2024  4.3058824 -4.421124 13.032886 0.7157016
## October 2024-August 2024   2.3960784 -5.942865 10.7350216 0.9624273
## November 2024-August 2024 -3.3227273 -11.787075  5.1416206 0.8690086
## December 2024-August 2024 -1.5829268 -10.113375  6.9475211 0.9947525
## January 2025-August 2024  -1.3119048 -9.819328  7.1955190 0.9978154
## October 2024-September 2024 -1.9098039 -7.664221  3.8446136 0.9316134
## November 2024-September 2024 -7.6286096 -13.563299 -1.6939207 0.0037255
## December 2024-September 2024 -5.8888092 -11.917398  0.1397801 0.0598467
## January 2025-September 2024 -5.6177871 -11.613753  0.3781787 0.0805726
```

```
## November 2024-October 2024 -5.7188057 -11.066503 -0.3711083 0.0283493
## December 2024-October 2024 -3.9790053 -9.430723 1.4727123 0.2919115
## January 2025-October 2024 -3.7079832 -9.123603 1.7076369 0.3638000
## December 2024-November 2024 1.7398004 -3.901869 7.3814700 0.9493916
## January 2025-November 2024 2.0108225 -3.595973 7.6176177 0.9070087
## January 2025-December 2024 0.2710221 -5.435072 5.9771161 0.9999933
```

```
## For one-way between subjects designs, partial eta squared is equivalent
## to eta squared. Returning eta squared.
```

```
## # Effect Size for ANOVA
##
## Parameter | Eta2 | 95% CI
## -----
## Month | 0.08 | [0.02, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

First off, a one-way ANOVA test of the two variables was performed (month and fluoro time). The test returned a significant result ($F(5, 218) = 3.877, p < 0.00219$) so there appears to be some significant difference between at least two of the months based on average fluoro time.

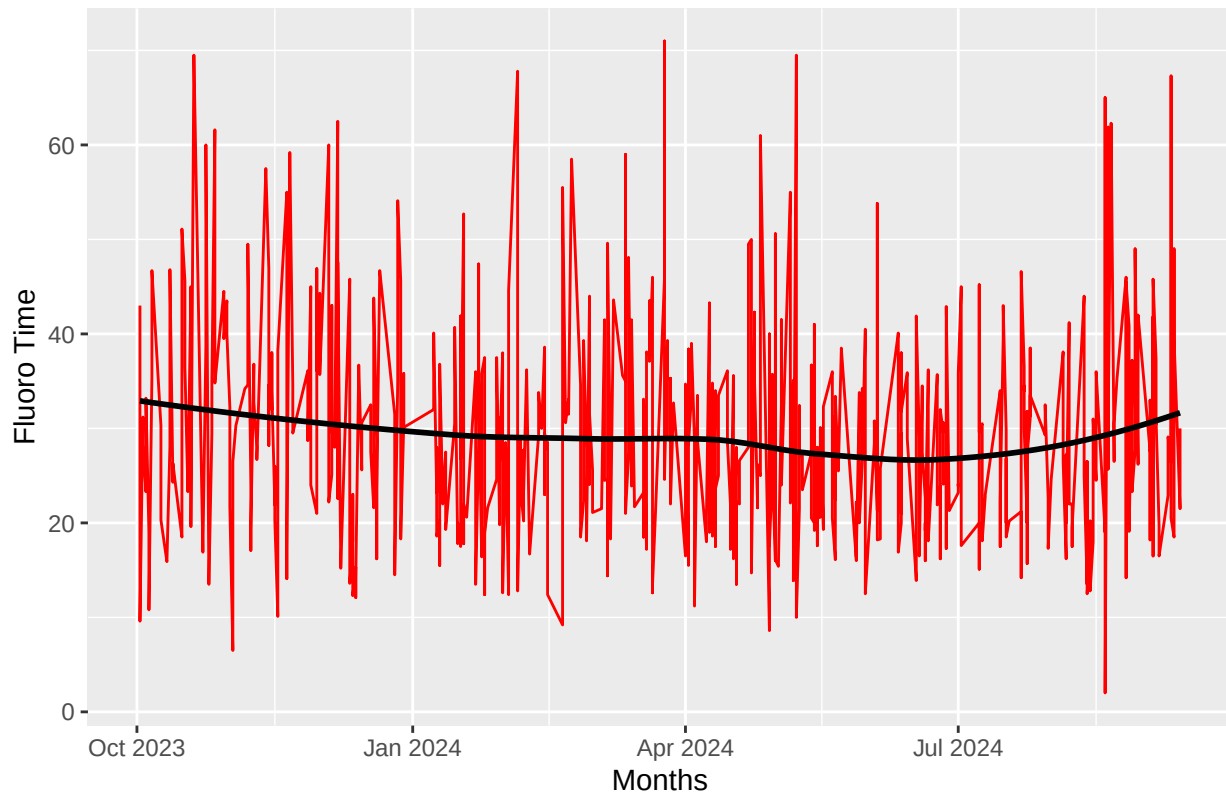
To properly identify the months in which there was a significant difference, the TukeyHSD post-hoc test was performed. This found a significant difference between October 2024 and November 2024, as well as September 2024 and November 2024. These p-values were 0.02 and 0.003 respectively ($p < 0.05$). No other month-to-month comparisons, including between mid-2024 and January 2025, reached statistical significance. This is notable because one might expect the greatest difference during this later period, potentially reflecting increased cumulative treatment volume or enhanced training.

Lastly, an effect size (Eta-squared) was calculated, which came out to 0.08, meaning 8% of the variation in fluoroscopy time can be explained by the month in which the procedure was performed. Conversely, 92% of the variation is due to variability in the month itself, which tells us other variables may be more of a factor (procedure volume, average age of the patient, systemic disease, general anatomy, etc.). It figures that the inherent anatomical variance of patients could vary significantly within a month, causing the bulk of the variation. So while there did appear to be a significant difference between the months (only Oct./Nov. 2024 and Sep./Nov. 2024) in terms of fluoroscopy time, only a small portion of the variance is attributed to the data between the months, but rather than within. That being said, 8% could be of significance in regards to physician performance given the correct clinical context.

Arizona Site

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Time Series of Arizona Fluro Time



Month	Mean_Fluoroscopy_Time
October 2023	33.2
November 2023	31.7
December 2023	30.4
January 2024	26.6
February 2024	29.3
March 2024	31.1
April 2024	28.2
May 2024	26.3
June 2024	26.7
July 2024	27.8
August 2024	29.1
September 2024	30.0

For Arizona, the time series shows a relatively straight trend line as well. The monthly averages are listed. We will test again for significance.

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Month      11   2294    208.6   1.579  0.101
## Residuals  529  69888    132.1
```

```
## Tukey multiple comparisons of means
```

```
##      95% family-wise confidence level
##
## Fit: aov(formula = Fluoroscopy_Time ~ Month, data = Arizona)
##
## $Month
```

	diff	lwr	upr	p adj
## November 2023-October 2023	-1.4853659	-9.925306	6.9545744	0.9999888
## December 2023-October 2023	-2.8441860	-11.187892	5.4995202	0.9937724
## January 2024-October 2023	-6.5666667	-14.821716	1.6883831	0.2758867
## February 2024-October 2023	-3.8880952	-12.278911	4.5027203	0.9343652
## March 2024-October 2023	-2.0860465	-10.429753	6.2576598	0.9996287
## April 2024-October 2023	-4.9650000	-12.726189	2.7961894	0.6229797
## May 2024-October 2023	-6.8839286	-14.753555	0.9856975	0.1542472
## June 2024-October 2023	-6.5066667	-14.761716	1.7483831	0.2894107
## July 2024-October 2023	-5.4234043	-13.596500	2.7496919	0.5661647
## August 2024-October 2023	-4.0777778	-12.007006	3.8514502	0.8731035
## September 2024-October 2023	-3.1692308	-12.722595	6.3841330	0.9950645
## December 2023-November 2023	-1.3588202	-9.595119	6.8774787	0.9999943
## January 2024-November 2023	-5.0813008	-13.227774	3.0651728	0.6603278
## February 2024-November 2023	-2.4027294	-10.686748	5.8812896	0.9985046
## March 2024-November 2023	-0.6006807	-8.836980	7.6356183	1.0000000
## April 2024-November 2023	-3.4796341	-11.125237	4.1659689	0.9418970
## May 2024-November 2023	-5.3985627	-13.154219	2.3570933	0.4885743
## June 2024-November 2023	-5.0213008	-13.167774	3.1251728	0.6772265
## July 2024-November 2023	-3.9380384	-12.001455	4.1253781	0.9075260
## August 2024-November 2023	-2.5924119	-10.408539	5.2237151	0.9950728
## September 2024-November 2023	-1.6838649	-11.143566	7.7758361	0.9999874
## January 2024-December 2023	-3.7224806	-11.769211	4.3242500	0.9350920
## February 2024-December 2023	-1.0439092	-9.229861	7.1420430	0.9999996
## March 2024-December 2023	0.7581395	-7.379518	8.8957966	1.0000000
## April 2024-December 2023	-2.1208140	-9.660051	5.4184226	0.9988713
## May 2024-December 2023	-4.0397425	-11.690562	3.6110772	0.8520681
## June 2024-December 2023	-3.6624806	-11.709211	4.3842500	0.9418659
## July 2024-December 2023	-2.5792182	-10.541851	5.3834150	0.9959870
## August 2024-December 2023	-1.2335917	-8.945705	6.4785211	0.9999958
## September 2024-December 2023	-0.3250447	-9.698987	9.0488972	1.0000000
## February 2024-January 2024	2.6785714	-5.416997	10.7741397	0.9951751
## March 2024-January 2024	4.4806202	-3.566110	12.5273508	0.8020134
## April 2024-January 2024	1.6016667	-5.839335	9.0426680	0.9999141
## May 2024-January 2024	-0.3172619	-7.871298	7.2367737	1.0000000
## June 2024-January 2024	0.0600000	-7.894765	8.0147650	1.0000000
## July 2024-January 2024	1.1432624	-6.726422	9.0129473	0.9999985
## August 2024-January 2024	2.4888889	-5.127219	10.1049965	0.9956660
## September 2024-January 2024	3.3974359	-5.897681	12.6925528	0.9888639
## March 2024-February 2024	1.8020487	-6.383903	9.9880009	0.9998927
## April 2024-February 2024	-1.0769048	-8.668245	6.5144350	0.9999988
## May 2024-February 2024	-2.9958333	-10.698001	4.7063348	0.9816899
## June 2024-February 2024	-2.6185714	-10.714140	5.4769968	0.9960362
## July 2024-February 2024	-1.5353090	-9.547292	6.4766744	0.9999733
## August 2024-February 2024	-0.1896825	-7.952738	7.5733732	1.0000000
## September 2024-February 2024	0.7188645	-8.697034	10.1347626	1.0000000
## April 2024-March 2024	-2.8789535	-10.418190	4.6602831	0.9841626
## May 2024-March 2024	-4.7978821	-12.448702	2.8529377	0.6525351
## June 2024-March 2024	-4.4206202	-12.467351	3.6261105	0.8157105

```
## July 2024-March 2024      -3.3373577 -11.299991  4.6252755 0.9676731
## August 2024-March 2024    -1.9917313  -9.703844  5.7203815 0.9994937
## September 2024-March 2024 -1.0831843 -10.457126  8.2907576 0.9999999
## May 2024-April 2024       -1.9189286  -8.929896  5.0920392 0.9991249
## June 2024-April 2024      -1.5416667  -8.982668  5.8993346 0.9999414
## July 2024-April 2024      -0.4584043  -7.808381  6.8915722 1.0000000
## August 2024-April 2024     0.8872222  -6.190582  7.9650262 0.9999997
## September 2024-April 2024  1.7957692  -7.063656 10.6551943 0.9999529
## June 2024-May 2024        0.3772619  -7.176774  7.9312975 1.0000000
## July 2024-May 2024        1.4605243  -6.003865  8.9249136 0.9999670
## August 2024-May 2024      2.8061508  -4.390394 10.0026961 0.9813296
## September 2024-May 2024   3.7146978  -5.239875 12.6692702 0.9700477
## July 2024-June 2024       1.0832624  -6.786422  8.9529473 0.9999991
## August 2024-June 2024     2.4288889  -5.187219 10.0449965 0.9964979
## September 2024-June 2024  3.3374359  -5.957681 12.6325528 0.9903785
## August 2024-July 2024     1.3456265  -6.181574  8.8728271 0.9999869
## September 2024-July 2024  2.2541735  -6.968237 11.4765838 0.9997013
## September 2024-August 2024 0.9085470  -8.098451  9.9155447 1.0000000
```

```
## For one-way between subjects designs, partial eta squared is equivalent
## to eta squared. Returning eta squared.
```

```
## # Effect Size for ANOVA
##
## Parameter | Eta2 |          95% CI
## -----
## Month     | 0.03 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

Another one-way ANOVA test of the two variables was performed. We have a p-value of 0.101 ($p < 0.05$) indicating no significant difference between the fluoro time between months over the year.

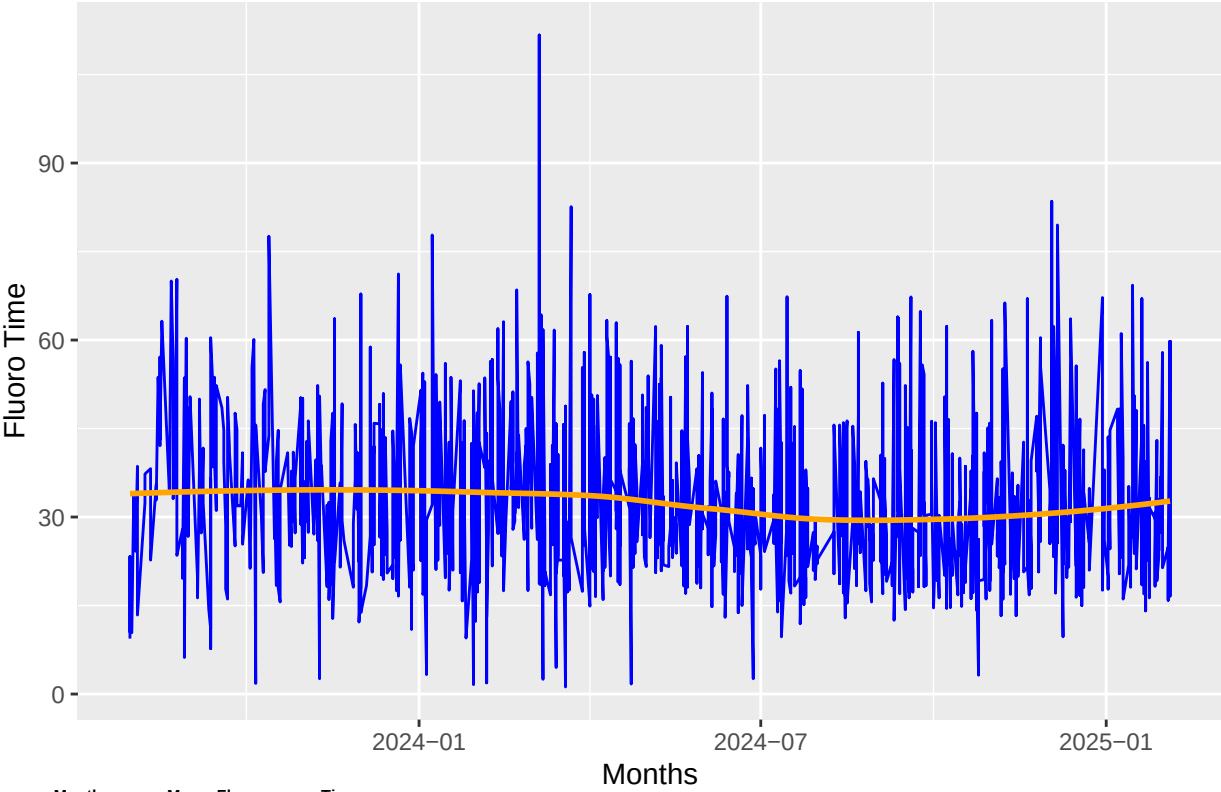
This was confirmed with the post-hoc test showing no significant difference between any monthly pairwise comparisons.

The effect size was calculated at 3% which is not statistically relevant in terms of explaining the variation of fluoroscopy time over the months. 3% is too small to be clinically relevant. In summary, Arizona did not show any meaningful difference in fluoroscopy time over the year of data analyzed.

St. Louis

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Time Series of St.Louis Fluro Time



Month	Mean_Fluoroscopy_Time
July 2023	14.4
August 2023	36.7
September 2023	35.1
October 2023	45.6
November 2023	32.1
December 2023	33.1
January 2024	32.8
February 2024	37.0
March 2024	47.8
April 2024	34.8
May 2024	43.3
June 2024	30.3
July 2024	29.2
August 2024	29.7
September 2024	32.0
October 2024	27.4
November 2024	30.4
December 2024	41.3
January 2025	31.7
February 2025	31.3

```

##           Df Sum Sq Mean Sq F value Pr(>F)
## Month      19   7850   413.1    2.237 0.00177 **
## Residuals 1069 197461   184.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Fluoroscopy_Time ~ Month, data = St.Louis)
##
## $Month
##           diff           lwr           upr           p adj
## August 2023-July 2023    22.76969697  -6.348215  51.8876085  0.3723346
## September 2023-July 2023  20.68055556  -8.336076  49.6971871  0.5589004
## October 2023-July 2023   21.18636364  -7.626640  49.9993672  0.4960769
## November 2023-July 2023  17.69814815 -10.944060  46.3403559  0.8030088
## December 2023-July 2023  18.69000000 -10.012444  47.3924438  0.7248635
## January 2024-July 2023   18.41129032 -10.133495  46.9560753  0.7402002
## February 2024-July 2023  22.63870968  -5.906075  51.1834946  0.3449946
## March 2024-July 2023     18.36666667 -10.235843  46.9691762  0.7471972
## April 2024-July 2023     20.37464789  -8.086514  48.8358099  0.5500774
## May 2024-July 2023       17.53661972 -10.924542  45.9977817  0.8067699
## June 2024-July 2023      15.86376812 -12.614111  44.3416474  0.9105551
## July 2024-July 2023      14.79142857 -13.677974  43.2608310  0.9520593
## August 2024-July 2023    15.30727273 -13.321227  43.9357728  0.9374716
## September 2024-July 2023  17.64202899 -10.835850  46.1199083  0.7995387
## October 2024-July 2023   13.03797468 -15.364705  41.4406548  0.9866557
## November 2024-July 2023  15.97936508 -12.554962  44.5136922  0.9065064
## December 2024-July 2023  18.25238095 -10.281946  46.7867080  0.7531744
## January 2025-July 2023   17.26338028 -11.197782  45.7245423  0.8267913
## February 2025-July 2023  16.85714286 -16.463772  50.1780581  0.9629282
## September 2023-August 2023 -2.08914141 -13.726191  9.5479084  1.0000000
## October 2023-August 2023  -1.58333333 -12.702919  9.5362528  1.0000000
## November 2023-August 2023 -5.07154882 -15.740761  5.5976632  0.9804587
## December 2023-August 2023 -4.07969697 -14.909577  6.7501830  0.9988313
## January 2024-August 2023  -4.35840665 -14.763250  6.0464370  0.9954356
## February 2024-August 2023 -0.13098729 -10.535831  10.2738563  1.0000000
## March 2024-August 2023    -4.40303030 -14.965207  6.1591463  0.9956982
## April 2024-August 2023    -2.39504908 -12.568238  7.7781395  0.9999991
## May 2024-August 2023      -5.23307725 -15.406266  4.9401113  0.9564089
## June 2024-August 2023     -6.90592885 -17.125793  3.3139357  0.6607904
## July 2024-August 2023     -7.97826840 -18.174488  2.2179515  0.3711065
## August 2024-August 2023   -7.46242424 -18.094782  3.1699337  0.5891510
## September 2024-August 2023 -5.12766798 -15.347533  5.0921966  0.9658440
## October 2024-August 2023  -9.73172229 -19.740132  0.2766872  0.0681694
## November 2024-August 2023 -6.79033189 -17.166451  3.5857870  0.7166344
## December 2024-August 2023 -4.51731602 -14.893435  5.8588029  0.9927380
## January 2025-August 2023  -5.50631669 -15.679505  4.6668719  0.9302469
## February 2025-August 2023 -5.91255411 -26.005822  14.1807138  0.9999672
## October 2023-September 2023 0.50580808 -10.345797  11.3574130  1.0000000
## November 2023-September 2023 -2.98240741 -13.372028  7.4072133  0.9999776
## December 2023-September 2023 -1.99055556 -12.545101  8.5639901  1.0000000
## January 2024-September 2023 -2.26926523 -12.387215  7.8486844  0.9999996

```

## February 2024-September 2023	1.95815412	-8.159795	12.0761037	1.0000000
## March 2024-September 2023	-2.31388889	-12.593563	7.9657857	0.9999996
## April 2024-September 2023	-0.30590767	-10.185477	9.5736621	1.0000000
## May 2024-September 2023	-3.14393584	-13.023506	6.7356340	0.9998926
## June 2024-September 2023	-4.81678744	-14.744414	5.1108389	0.9757336
## July 2024-September 2023	-5.88912698	-15.792411	4.0141570	0.8499556
## August 2024-September 2023	-5.37328283	-15.725054	4.9784885	0.9524826
## September 2024-September 2023	-3.03852657	-12.966153	6.8890998	0.9999401
## October 2024-September 2023	-7.64258087	-17.352390	2.0672283	0.3596217
## November 2024-September 2023	-4.70119048	-14.789599	5.3872176	0.9842400
## December 2024-September 2023	-2.42817460	-12.516583	7.6602335	0.9999987
## January 2025-September 2023	-3.41717527	-13.296745	6.4623945	0.9996399
## February 2025-September 2023	-3.82341270	-23.769629	16.1228037	1.0000000
## November 2023-October 2023	-3.48821549	-13.294769	6.3183377	0.9994663
## December 2023-October 2023	-2.49636364	-12.477481	7.4847536	0.9999975
## January 2024-October 2023	-2.77507331	-12.293328	6.7431813	0.9999715
## February 2024-October 2023	1.45234604	-8.065909	10.9706007	1.0000000
## March 2024-October 2023	-2.81969697	-12.509691	6.8702966	0.9999724
## April 2024-October 2023	-0.81171575	-10.076173	8.4527414	1.0000000
## May 2024-October 2023	-3.64974392	-12.914201	5.6147132	0.9978935
## June 2024-October 2023	-5.32259552	-14.638283	3.9930919	0.8898050
## July 2024-October 2023	-6.39493506	-15.684677	2.8948066	0.6264029
## August 2024-October 2023	-5.87909091	-15.645535	3.8873535	0.8359513
## September 2024-October 2023	-3.54433465	-12.860022	5.7713528	0.9986656
## October 2024-October 2023	-8.14838895	-17.231597	0.9348187	0.1467244
## November 2024-October 2023	-5.20699856	-14.693844	4.2798473	0.9211805
## December 2024-October 2023	-2.93398268	-12.420829	6.5528632	0.9999299
## January 2025-October 2023	-3.92298335	-13.187441	5.3414738	0.9947908
## February 2025-October 2023	-4.32922078	-23.978033	15.3195913	0.9999997
## December 2023-November 2023	0.99185185	-8.484940	10.4686440	1.0000000
## January 2024-November 2023	0.71314217	-8.274853	9.7011375	1.0000000
## February 2024-November 2023	4.94056153	-4.047434	13.9285569	0.9201582
## March 2024-November 2023	0.66851852	-8.501152	9.8381895	1.0000000
## April 2024-November 2023	2.67649974	-6.042276	11.3952758	0.9999374
## May 2024-November 2023	-0.16152843	-8.880304	8.5572476	1.0000000
## June 2024-November 2023	-1.83438003	-10.607573	6.9388133	0.9999999
## July 2024-November 2023	-2.90671958	-11.652358	5.8389187	0.9997967
## August 2024-November 2023	-2.39087542	-11.641298	6.8595476	0.9999958
## September 2024-November 2023	-0.05611916	-8.829312	8.7170742	1.0000000
## October 2024-November 2023	-4.66017346	-13.186107	3.8657605	0.9239744
## November 2024-November 2023	-1.71878307	-10.673510	7.2359438	1.0000000
## December 2024-November 2023	0.55423280	-8.400494	9.5089597	1.0000000
## January 2025-November 2023	-0.43476787	-9.153544	8.2840082	1.0000000
## February 2025-November 2023	-0.84100529	-20.238498	18.5564870	1.0000000
## January 2024-December 2023	-0.27870968	-9.456851	8.8994315	1.0000000
## February 2024-December 2023	3.94870968	-5.229432	13.1268509	0.9936915
## March 2024-December 2023	-0.32333333	-9.679459	9.0327919	1.0000000
## April 2024-December 2023	1.68464789	-7.230018	10.5993139	1.0000000
## May 2024-December 2023	-1.15338028	-10.068046	7.7612858	1.0000000
## June 2024-December 2023	-2.82623188	-11.794127	6.1416629	0.9999071
## July 2024-December 2023	-3.89857143	-12.839511	5.0423684	0.9926032
## August 2024-December 2023	-3.38272727	-12.818009	6.0525544	0.9994045
## September 2024-December 2023	-1.04797101	-10.015866	7.9199237	1.0000000
## October 2024-December 2023	-5.65202532	-14.378179	3.0741288	0.7334928

## November 2024-December 2023	-2.71063492	-11.856199	6.4349295	0.9999633
## December 2024-December 2023	-0.43761905	-9.583183	8.7079454	1.0000000
## January 2025-December 2023	-1.42661972	-10.341286	7.4880463	1.0000000
## February 2025-December 2023	-1.83285714	-21.319184	17.6534694	1.0000000
## February 2024-January 2024	4.22741935	-4.445109	12.8999476	0.9745332
## March 2024-January 2024	-0.04462366	-8.905298	8.8160507	1.0000000
## April 2024-January 2024	1.96335756	-6.429839	10.3565537	0.9999992
## May 2024-January 2024	-0.87467060	-9.267867	7.5185255	1.0000000
## June 2024-January 2024	-2.54752221	-10.997233	5.9021882	0.9999524
## July 2024-January 2024	-3.61986175	-12.040959	4.8012353	0.9937567
## August 2024-January 2024	-3.10401760	-12.048234	5.8401991	0.9996224
## September 2024-January 2024	-0.76926134	-9.218972	7.6804491	1.0000000
## October 2024-January 2024	-5.37331564	-13.566010	2.8193784	0.7129467
## November 2024-January 2024	-2.43192524	-11.069970	6.2061196	0.9999835
## December 2024-January 2024	-0.15890937	-8.796954	8.4791355	1.0000000
## January 2025-January 2024	-1.14791004	-9.541106	7.2452861	1.0000000
## February 2025-January 2024	-1.55414747	-20.807495	17.6992001	1.0000000
## March 2024-February 2024	-4.27204301	-13.132717	4.5886314	0.9772862
## April 2024-February 2024	-2.26406179	-10.657258	6.1291343	0.9999916
## May 2024-February 2024	-5.10208996	-13.495286	3.2911062	0.8240975
## June 2024-February 2024	-6.77494156	-15.224652	1.6747689	0.3243126
## July 2024-February 2024	-7.84728111	-16.268378	0.5738159	0.1044695
## August 2024-February 2024	-7.33143695	-16.275654	1.6127797	0.2839728
## September 2024-February 2024	-4.99668069	-13.446391	3.4530298	0.8560867
## October 2024-February 2024	-9.60073499	-17.793429	-1.4080410	0.0053641
## November 2024-February 2024	-6.65934460	-15.297389	1.9787003	0.4004645
## December 2024-February 2024	-4.38632873	-13.024374	4.2517162	0.9615464
## January 2025-February 2024	-5.37532940	-13.768526	3.0178667	0.7513062
## February 2025-February 2024	-5.78156682	-25.034914	13.4717807	0.9999552
## April 2024-March 2024	2.00798122	-6.579483	10.5954457	0.9999992
## May 2024-March 2024	-0.83004695	-9.417511	7.7574175	1.0000000
## June 2024-March 2024	-2.50289855	-11.145607	6.1398100	0.9999743
## July 2024-March 2024	-3.57523810	-12.189974	5.0394981	0.9959284
## August 2024-March 2024	-3.05939394	-12.186158	6.0673698	0.9997700
## September 2024-March 2024	-0.72463768	-9.367346	7.9180709	1.0000000
## October 2024-March 2024	-5.32869198	-13.720297	3.0629133	0.7642833
## November 2024-March 2024	-2.38730159	-11.214228	6.4396245	0.9999912
## December 2024-March 2024	-0.11428571	-8.941212	8.7126404	1.0000000
## January 2025-March 2024	-1.10328638	-9.690751	7.4841781	1.0000000
## February 2025-March 2024	-1.50952381	-20.848350	17.8293023	1.0000000
## May 2024-April 2024	-2.83802817	-10.942270	5.2662137	0.9995711
## June 2024-April 2024	-4.51087977	-12.673637	3.6518772	0.9164254
## July 2024-April 2024	-5.58321932	-13.716353	2.5499148	0.6316348
## August 2024-April 2024	-5.06737516	-13.741014	3.6062634	0.8687220
## September 2024-April 2024	-2.73261890	-10.895376	5.4301381	0.9997744
## October 2024-April 2024	-7.33667320	-15.233079	0.5597330	0.1074065
## November 2024-April 2024	-4.39528281	-12.752843	3.9622775	0.9462765
## December 2024-April 2024	-2.12226693	-10.479827	6.2352934	0.9999968
## January 2025-April 2024	-3.11126761	-11.215509	4.9929742	0.9984969
## February 2025-April 2024	-3.51750503	-22.646655	15.6116451	1.0000000
## June 2024-May 2024	-1.67285160	-9.835609	6.4899054	0.9999999
## July 2024-May 2024	-2.74519115	-10.878325	5.3879429	0.9997460
## August 2024-May 2024	-2.22934699	-10.902986	6.4442916	0.9999961
## September 2024-May 2024	0.10540927	-8.057348	8.2681663	1.0000000

```

## October 2024-May 2024      -4.49864503 -12.395051  3.3977611 0.8923433
## November 2024-May 2024     -1.55725464 -9.914815  6.8003057 1.0000000
## December 2024-May 2024      0.71576123 -7.641799  9.0733215 1.0000000
## January 2025-May 2024       -0.27323944 -8.377481  7.8310024 1.0000000
## February 2025-May 2024      -0.67947686 -19.808627 18.4496733 1.0000000
## July 2024-June 2024         -1.07233954 -9.263782  7.1191033 1.0000000
## August 2024-June 2024       -0.55649539 -9.284833  8.1718419 1.0000000
## September 2024-June 2024    1.77826087 -6.442595  9.9991165 0.9999998
## October 2024-June 2024      -2.82579343 -10.782243  5.1306565 0.9994776
## November 2024-June 2024     0.11559696 -8.298717  8.5299110 1.0000000
## December 2024-June 2024     2.38861284 -6.025701 10.8029268 0.9999812
## January 2025-June 2024      1.39961217 -6.763145  9.5623692 1.0000000
## February 2025-June 2024     0.99337474 -18.160639 20.1473887 1.0000000
## August 2024-July 2024       0.51584416 -8.184796  9.2164844 1.0000000
## September 2024-July 2024    2.85060041 -5.340842 11.0420432 0.9996074
## October 2024-July 2024      -1.75345389 -9.679510  6.1726021 0.9999997
## November 2024-July 2024     1.18793651 -7.197643  9.5735163 1.0000000
## December 2024-July 2024     3.46095238 -4.924627 11.8465322 0.9961980
## January 2025-July 2024      2.47195171 -5.661182 10.6050858 0.9999461
## February 2025-July 2024     2.06571429 -17.075694 21.2071228 1.0000000
## September 2024-August 2024  2.33475626 -6.393581 11.0630936 0.9999927
## October 2024-August 2024    -2.26929804 -10.749068  6.2104721 0.9999926
## November 2024-August 2024   0.67209235 -8.238692  9.5828771 1.0000000
## December 2024-August 2024   2.94510823 -5.965677 11.8558930 0.9998126
## January 2025-August 2024    1.95610755 -6.717531 10.6297462 0.9999996
## February 2025-August 2024   1.54987013 -17.827376 20.9271160 1.0000000
## October 2024-September 2024 -4.60405430 -12.560504  3.3523957 0.8781284
## November 2024-September 2024 -1.66266391 -10.076978  6.7516501 1.0000000
## December 2024-September 2024 0.61035197 -7.803962  9.0246660 1.0000000
## January 2025-September 2024 -0.37864870 -8.541406  7.7841083 1.0000000
## February 2025-September 2024 -0.78488613 -19.938900 18.3691279 1.0000000
## November 2024-October 2024  2.94139040 -5.214792 11.0975726 0.9993540
## December 2024-October 2024  5.21440627 -2.941776 13.3705885 0.7540029
## January 2025-October 2024    4.22540560 -3.671001 12.1218118 0.9370281
## February 2025-October 2024   3.81916817 -15.222861 22.8611975 0.9999999
## December 2024-November 2024  2.27301587 -6.330407 10.8764392 0.9999940
## January 2025-November 2024   1.28401520 -7.073545  9.6415755 1.0000000
## February 2025-November 2024  0.87777778 -18.360062 20.1156172 1.0000000
## January 2025-December 2024  -0.98900067 -9.346561  7.3685596 1.0000000
## February 2025-December 2024 -1.39523810 -20.633077 17.8426013 1.0000000
## February 2025-January 2025  -0.40623742 -19.535388 18.7229127 1.0000000

```

```

## For one-way between subjects designs, partial eta squared is equivalent
## to eta squared. Returning eta squared.

```

```

## # Effect Size for ANOVA
##
## Parameter | Eta2 |          95% CI
## -----
## Month     | 0.04 | [0.01, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].

```

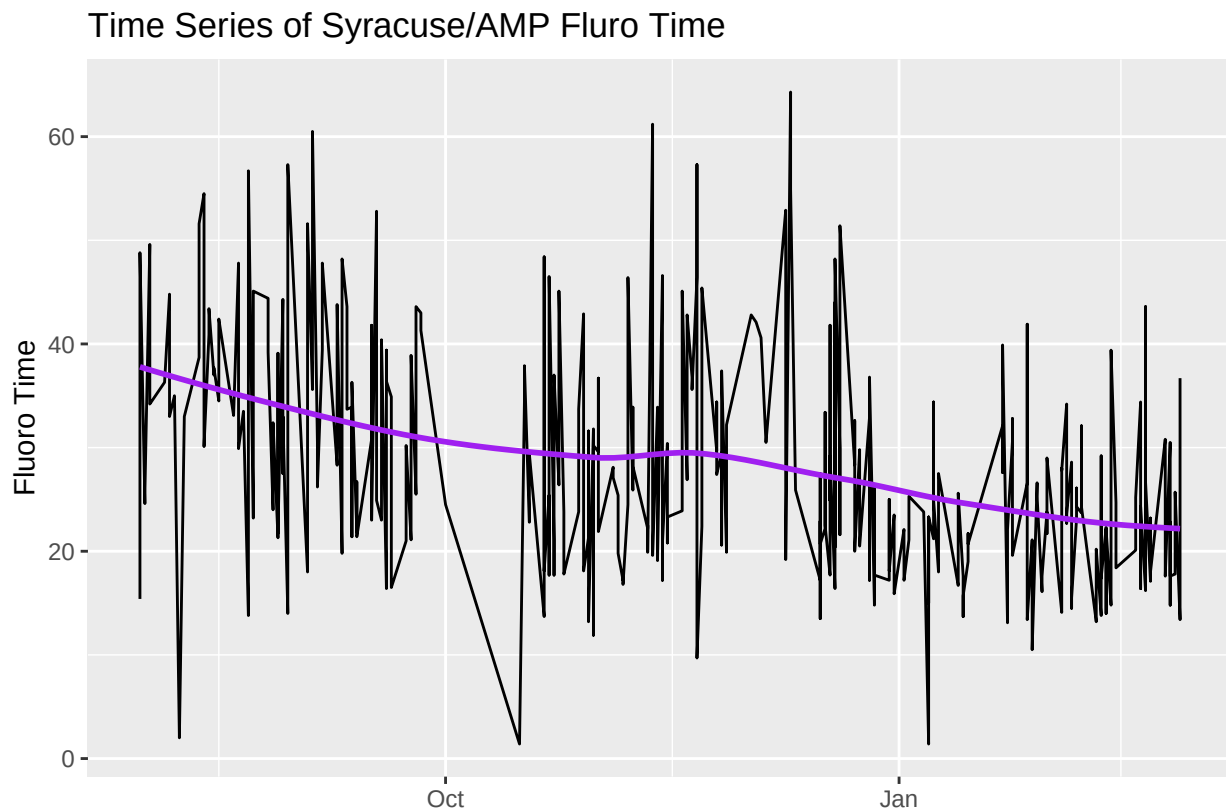
For St. Louis, the ANOVA returned a significant effect of month on fluoroscopy time ($p = 0.001440$, $p < 0.05$).

However, the post-hoc test (again using Tukey's HSD) revealed no significant pairwise interactions between the individual months, and the effect size remains extremely low (0.04). It would be hard to attribute any change in fluoroscopy time to monthly performance, and to that affect, the monthly change in performance of the provider.

Taken together, the ANOVA indicated some statistically significant effect of month on fluoroscopy time, but the absence of any significant pairwise comparisons, and small effect size indicate that the data is practically meaningful. It would not be appropriate to attribute the differences to physician improvement.

Syracuse/AMP

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Month	Mean_Fluoroscopy_Time
July 2024	32.1
August 2024	35.1
September 2024	33.5
October 2024	26.9
November 2024	29.2
December 2024	28.4
January 2025	22.2
February 2025	23.0

```
## # A tibble: 8 x 2
##   Month      Count
##   <chr>      <int>
## 1 July 2024      2
## 2 August 2024   41
## 3 September 2024 48
## 4 October 2024  33
## 5 November 2024 54
## 6 December 2024 49
## 7 January 2025  46
## 8 February 2025 57
```

Interestingly enough the trend line for Syracuse does appear to be decreasing in a meaningful way. There is only 2 cases for July of 2024, but there is > 28 for all other months which satisfies the appropriate case volume when generating the monthly averages.

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Month      7   6720    960.0    9.487 1.03e-10 ***
## Residuals 322  32585    101.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## For one-way between subjects designs, partial eta squared is equivalent
##   to eta squared. Returning eta squared.

## # Effect Size for ANOVA
##
## Parameter | Eta2 |      95% CI
## -----
## Month     | 0.17 | [0.10, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

```
##           diff      lwr      upr      p adj
## October 2024-August 2024 -8.308647 -15.48638 -1.1309180 1.103740e-02
## December 2024-August 2024 -6.794176 -13.29026 -0.2980955 3.310251e-02
## January 2025-August 2024 -12.952916 -19.54479 -6.3610421 1.518969e-07
## February 2025-August 2024 -12.211040 -18.49602 -5.9260612 2.200962e-07
## January 2025-September 2024 -11.338587 -17.67122 -5.0059511 2.599161e-06
## February 2025-September 2024 -10.596711 -16.60923 -4.5841879 4.023862e-06
## January 2025-November 2024 -6.939050 -13.09712 -0.7809841 1.512797e-02
## February 2025-November 2024 -6.197173 -12.02555 -0.3688009 2.814210e-02
```

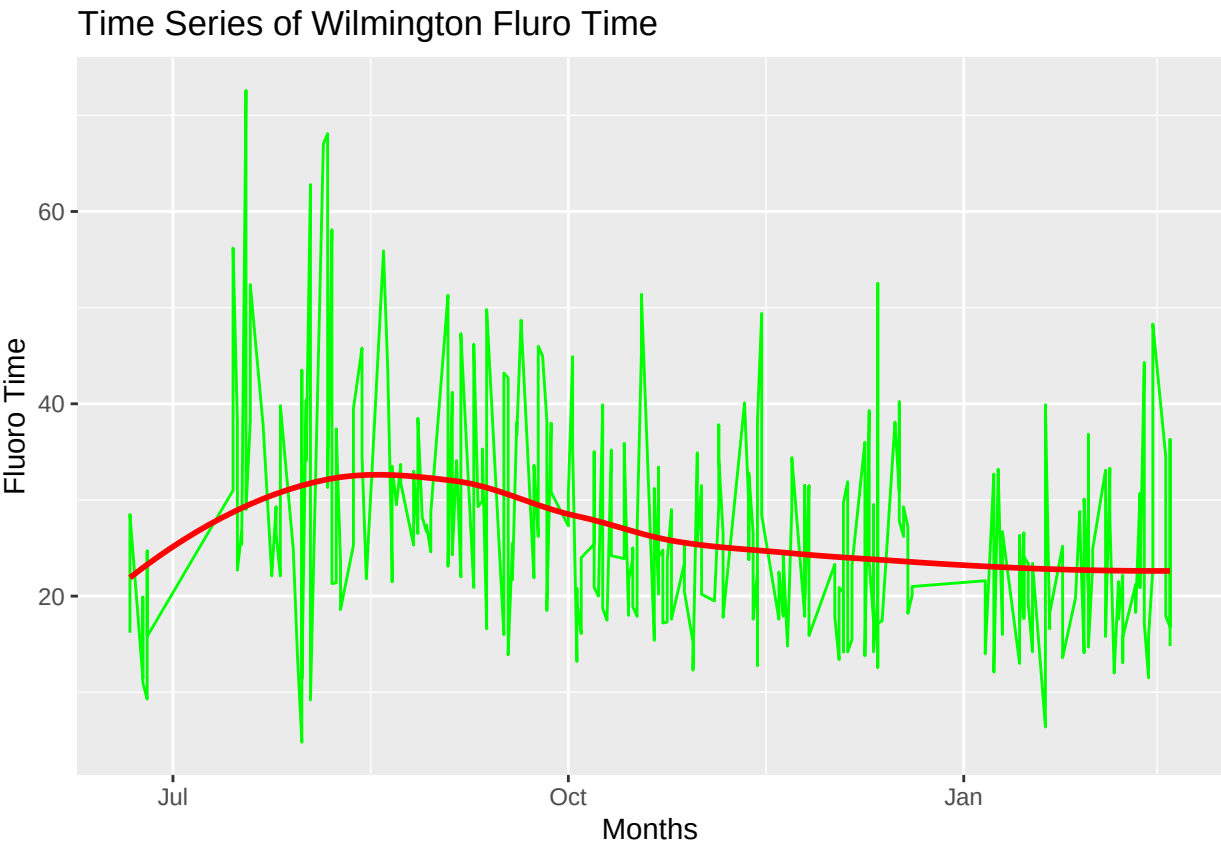
As expected, the ANOVA test returned a very low p-value of 5.1×10^{-10} ($p < 0.05$, 95% CI), confirming that there is a significant change in average fluoroscopy time over the course of the given time frame. Given the extensive list of pairwise comparisons, they were condensed to show the months in which there is significant pairwise interaction. While there is some significant month to month variation, it's important to note that there is also significant variation across January/February 2025 and August 2024. This allows the inference that the provider significantly changed his average fluoroscopy time over the entire time frame. The effect size of 18% indicates that the month of the procedure explains about 18% of the variability in fluoroscopy time. This suggests a moderate effect, likely reflecting the provider's improvement over time. However, this (and all prior) analysis considers only the passage of time (months) and does not account for other potential

factors that might also influence fluoroscopy time, which in the scope of interventional radiology, can be quite large.

In summary, the 18% effect size reflects how much of the variability in fluoroscopy time is explained by time progression (months). Skill improvement is probably a major contributor to this, but the effect also includes all other time-related changes.

Wilmington

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Month	Mean_Fluoroscopy_Time
June 2024	16.4
July 2024	31.8
August 2024	34.0
September 2024	32.8
October 2024	25.2
November 2024	26.1
December 2024	24.1
January 2025	22.2
February 2025	22.8

```
## # A tibble: 9 x 2
```

```

##      Month      Count
##      <chr>      <int>
## 1 June 2024      9
## 2 July 2024     21
## 3 August 2024    35
## 4 September 2024 33
## 5 October 2024   48
## 6 November 2024  32
## 7 December 2024  35
## 8 January 2025   36
## 9 February 2025  28

##              Df Sum Sq Mean Sq F value    Pr(>F)
## Month          8   6147    768.4    7.051 1.88e-08 ***
## Residuals    268  29206    109.0
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## For one-way between subjects designs, partial eta squared is equivalent
## to eta squared. Returning eta squared.

## # Effect Size for ANOVA
##
## Parameter | Eta2 |          95% CI
## -----
## Month      | 0.17 | [0.09, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].

##              diff      lwr      upr      p adj
## July 2024-June 2024    15.404762  2.397851 28.4116724 0.0078103765
## August 2024-June 2024  17.625714  5.424151 29.8272779 0.0003183867
## September 2024-June 2024 16.442424  4.165464 28.7193844 0.0012518379
## January 2025-July 2024  -9.607540 -18.571934 -0.6431453 0.0253878857
## October 2024-August 2024 -8.875714 -16.132242 -1.6191870 0.0050477833
## December 2024-August 2024 -9.940000 -17.744146 -2.1358537 0.0027950519
## January 2025-August 2024 -11.828492 -19.578253 -4.0787307 0.0001030455
## February 2025-August 2024 -11.225714 -19.503261 -2.9481671 0.0010117476
## October 2024-September 2024 -7.692424 -15.075024 -0.3098241 0.0338789598
## December 2024-September 2024 -8.756710 -16.678218 -0.8352015 0.0180992410
## January 2025-September 2024 -10.645202 -18.513137 -2.7772672 0.0010533648
## February 2025-September 2024 -10.042424 -18.430713 -1.6541358 0.0067704539

```

The ANOVA for Wilmington returned a p-value of 1.88×10^{-8} , showing high significance in fluoro time differences between the months ($p < 0.05$, 95% CI). The effect size was 17%, similar to Syracuse. See the explanation under the Syracuse analysis, as it would be practically identical to what would be written here. There again was a significant difference between the earliest month and the last months of the data set

CROSS SITE ANALYSIS

Generalized Additive Model

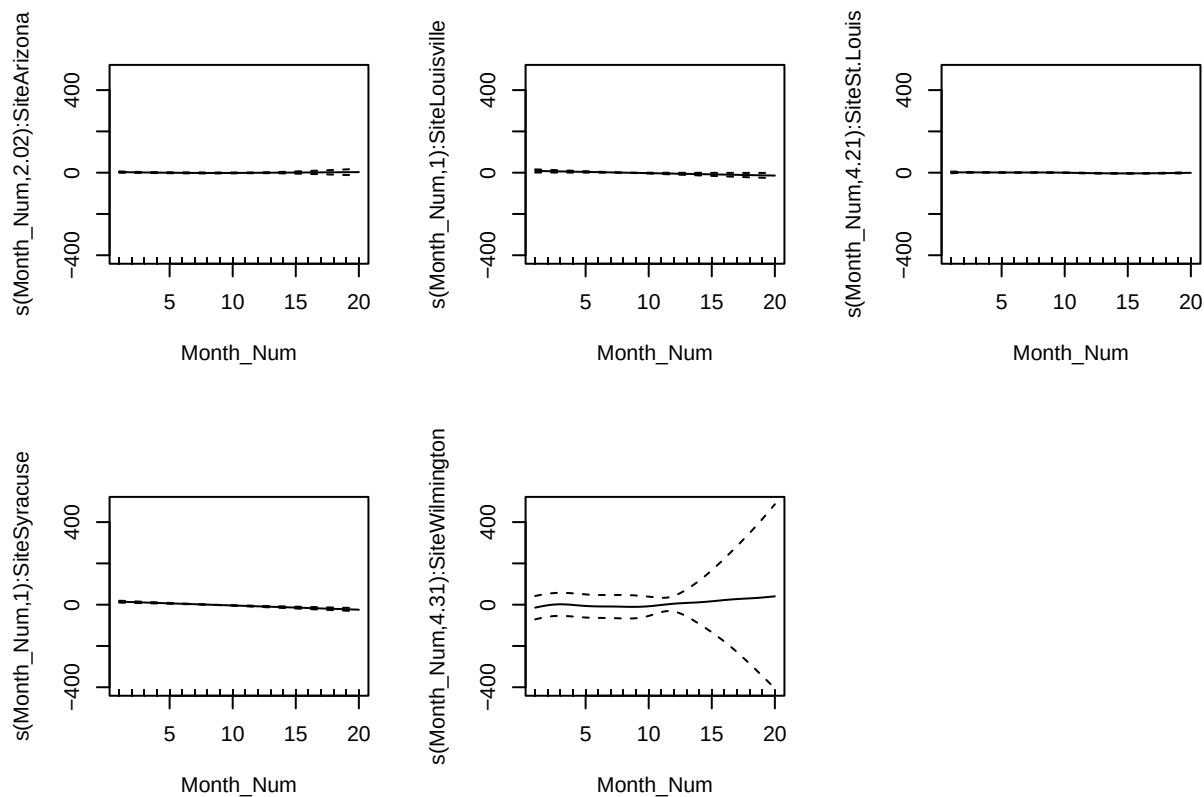
Now that each site has been analyzed individually, the next step is to evaluate which training method performs best across all clinical sites collectively. To analyze the variation in fluoroscopy time over several

months across multiple sites, I chose to use a Generalized Additive Model (GAM) with Site included as a factor variable.

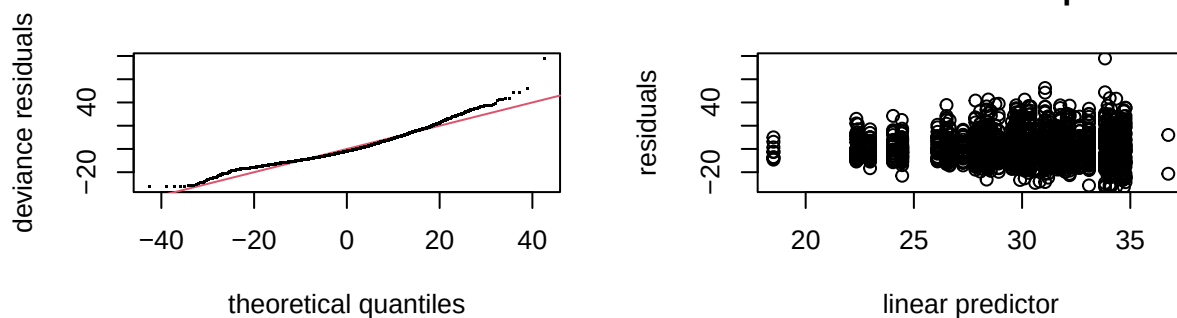
This modeling approach offers the flexibility to capture nonlinear trends in fluoroscopy time over time within each site, while explicitly accounting for differences between sites as categorical groups. Treating Site as a factor ensures the model recognizes each location as a distinct entity and estimates unique smooth temporal patterns for each site, rather than assuming a uniform or numeric relationship across all sites.

This flexibility is important because each site may have different training methods, workflows, or operational practices that produce distinct temporal dynamics in fluoroscopy times. By including site-specific smooth terms, the GAM can more accurately model these complex patterns, enabling a deeper understanding of how fluoroscopy efficiency evolves differently across locations and, ultimately, how the various training methods compare.

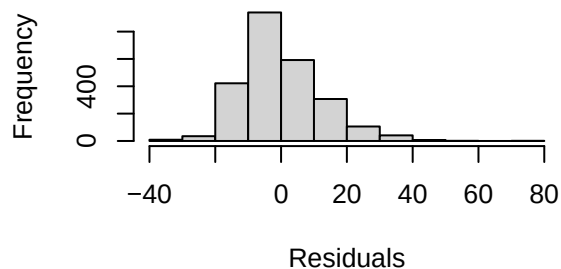
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## Fluoroscopy_Time ~ Site + s(Month_Num, by = Site)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   29.281     1.004   29.175 < 2e-16 ***
## SiteLouisville -5.685     2.625   -2.166  0.03044 *
## SiteSt.Louis   3.943     1.099    3.588  0.00034 ***
## SiteSyracuse   -7.113     1.542   -4.613  4.17e-06 ***
## SiteWilmington  3.623    27.961    0.130  0.89692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(Month_Num):SiteArizona    2.020  2.507  2.230 0.057419 .
## s(Month_Num):SiteLouisville 1.000  1.000  4.802 0.028527 *
## s(Month_Num):SiteSt.Louis   4.211  5.181  4.847 0.000174 ***
## s(Month_Num):SiteSyracuse   1.000  1.000 39.532 < 2e-16 ***
## s(Month_Num):SiteWilmington 4.307  4.661  6.407 8.78e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.0656   Deviance explained = 7.19%
## GCV = 145.93   Scale est. = 144.89      n = 2461
```



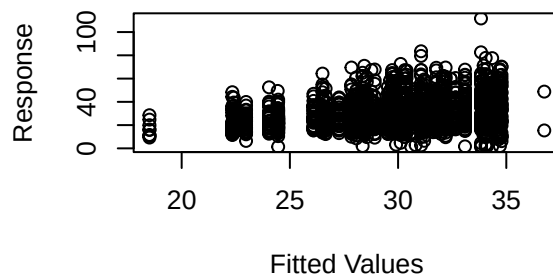
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 11 iterations.
## The RMS GCV score gradient at convergence was 3.98953e-06 .
```

```

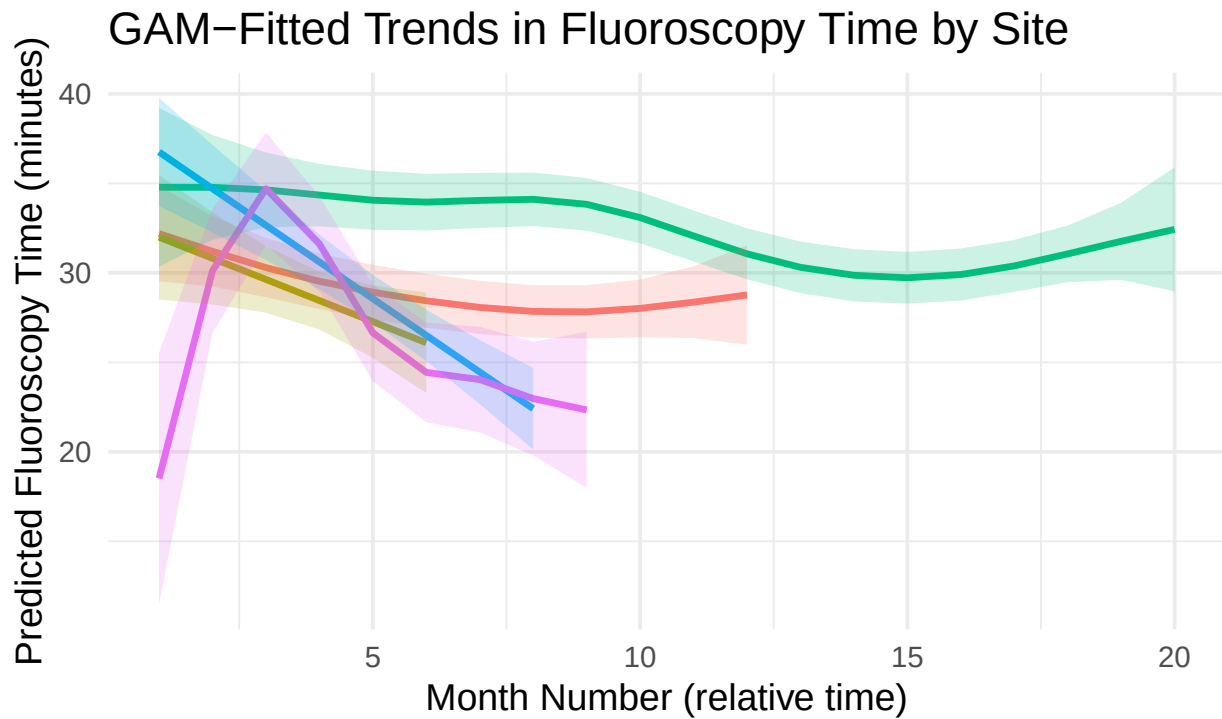
## The Hessian was positive definite.
## Model rank = 50 / 50
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'   edf k-index p-value
## s(Month_Num):SiteArizona    9.00 2.02    1.03    0.95
## s(Month_Num):SiteLouisville 9.00 1.00    1.03    0.94
## s(Month_Num):SiteSt.Louis   9.00 4.21    1.03    0.94
## s(Month_Num):SiteSyracuse   9.00 1.00    1.03    0.94
## s(Month_Num):SiteWilmington 9.00 4.31    1.03    0.95

```

The parametric coefficients indicate significant baseline differences in average fluoroscopy time between sites. Specifically, compared to the reference site (Arizona), Louisville and Syracuse had significantly lower baseline fluoroscopy times (approximately 5.7 and 7 units lower, respectively; $p < 0.05$), while St. Louis had a significantly higher baseline time (+4 units, $p < 0.001$). Wilmington did not differ significantly from Arizona in baseline fluoroscopy time.

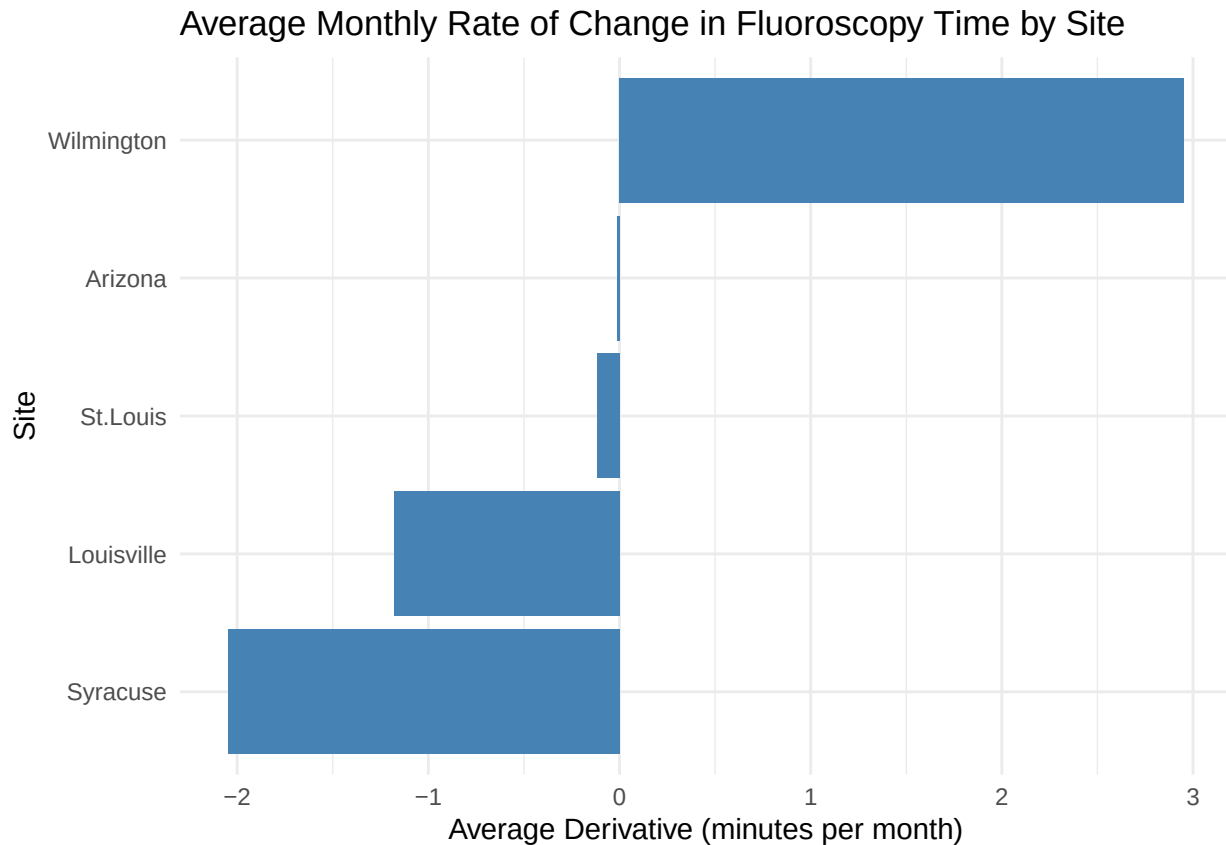
The smooth terms for month within each site revealed significant nonlinear temporal patterns for most sites. St. Louis, Syracuse, and Wilmington exhibited complex and highly significant nonlinear trends (effective degrees of freedom > 4 , $p < 0.001$), indicating that fluoroscopy times changed in a non-monotonic way across months. Louisville's temporal trend was simpler but still significant (edf = 1, $p = 0.028$), while Arizona's trend was marginally nonsignificant ($p = 0.056$). These findings suggest that fluoroscopy time performance evolved differently over time at each site, with some sites showing periods of improvement or fluctuation not captured by simple linear trends.

Overall, this model demonstrates meaningful differences in fluoroscopy time between sites and reveals distinct temporal trajectories within each site. This supports the hypothesis that site-specific factors, such as training methods and operational practices, influence the evolution of fluoroscopy efficiency over time. The GAM's flexibility in capturing nonlinear time effects provides a nuanced understanding that simpler linear models would miss.



The figure displays site-specific temporal trends in fluoroscopy time as modeled by the GAM. Solid lines represent the predicted mean fluoroscopy time across months for each site, while the shaded ribbons indicate approximate 95% confidence intervals around these predictions. The varying shapes of the curves highlight differences in how fluoroscopy time evolved over time at each clinical site. Sites with more pronounced curves and wider ribbons reflect greater nonlinear changes and uncertainty in the trend, consistent with the GAM results. This visualization complements the statistical findings by illustrating the unique temporal patterns captured by the model.

```
## # A tibble: 5 x 4
##   Site      Avg_Derivative Min_Derivative Max_Derivative
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 Syracuse      -2.05          -2.05          -2.05
## 2 Louisville    -1.18          -1.18          -1.18
## 3 St.Louis      -0.119         -1.06           0.710
## 4 Arizona       -0.0138        -1.00           0.413
## 5 Wilmington     2.95          -5.33          13.0
```



To better understand how fluoroscopy time is evolving over the study period at each clinical site, the first derivatives of the smooth terms from the GAM model were computed. These derivatives represent the instantaneous rate of change in fluoroscopy time with respect to month number. Negative derivative values indicate a decrease in fluoroscopy time (improvement), while positive values indicate an increase (worsening).

The derivatives were summarized by site, calculating the average rate of change as well as the minimum and maximum observed rates during the study period. This summary provides insight into both the overall trend and the variability of change over time for each site.

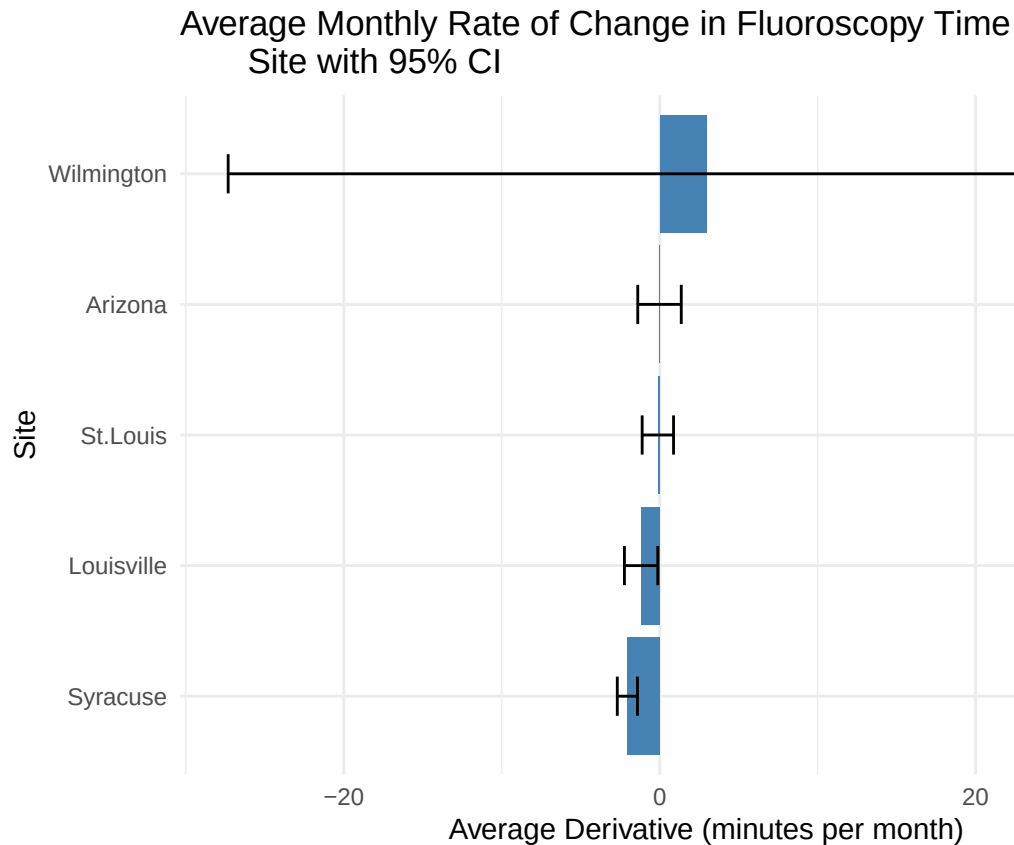
Louisville and Syracuse show a negative average derivative indicating improvement in fluoroscopy time over time

Arizona and St. Louis show a relatively neutral average derivative over time indicating fluctuating or stable performance

Wilmington showed an average derivative increase over time, concluding that the fluoroscopy time actually worsened as the time period went on

The accompanying bar plot visualizes these average rates of change for each site, with error bars representing the range of observed rates. This allows for quick comparison of performance dynamics across sites and highlights those with the most favorable or concerning trends.

Overall, this derivative analysis complements the GAM-fitted trends by quantifying the speed and direction of change, helping to identify which sites are improving fluoroscopy efficiency most rapidly, which are stable, and

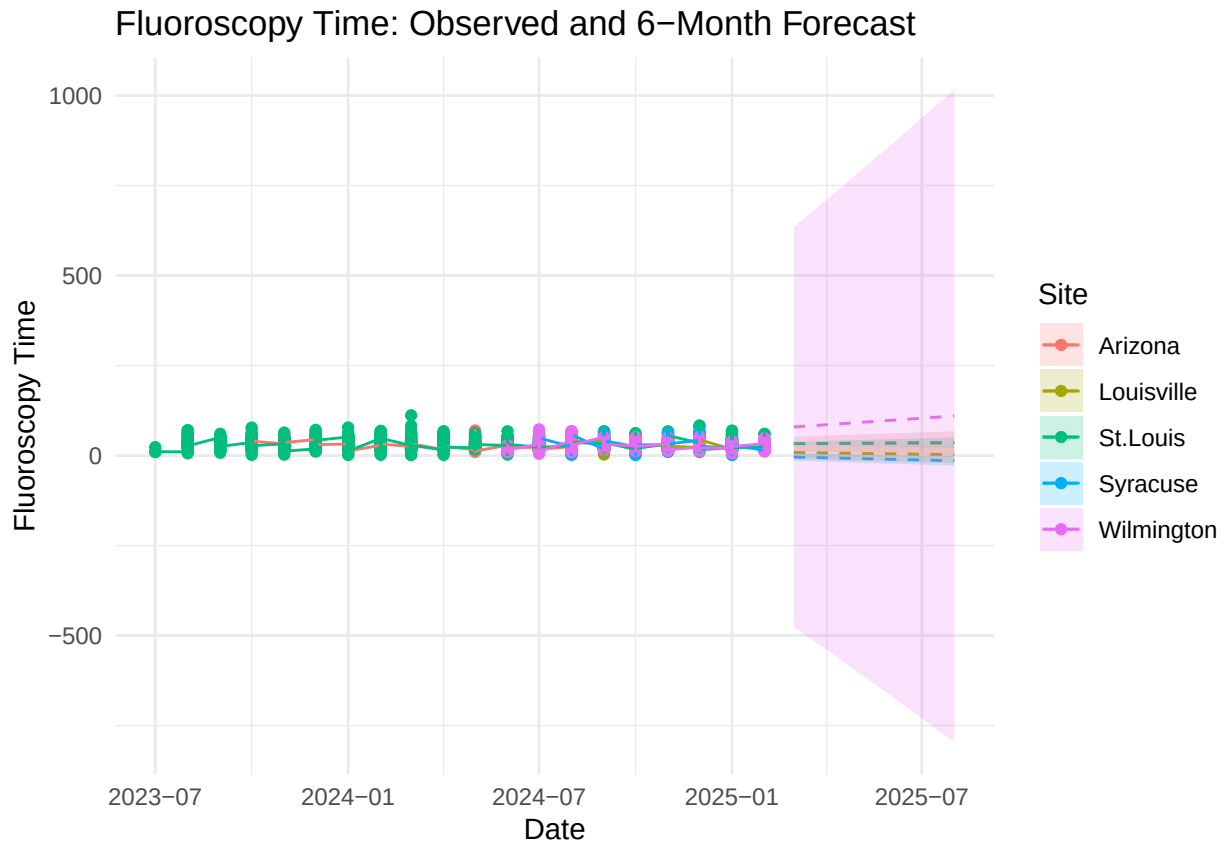


which may need further investigation.

The bar plot displays the average monthly rate of change for each site, with error bars representing the 95% confidence intervals. Sites with confidence intervals entirely below zero suggest statistically significant decreases in fluoroscopy time, indicating consistent improvements. Conversely, sites with confidence intervals overlapping zero suggest less certain or more variable trends.

This analysis offers a clear comparative view of how fluoroscopy efficiency evolves across sites, highlighting those with robust improvements and others where changes may be less definitive. Such insights can guide targeted quality improvement efforts tailored to each site's unique performance trajectory.

The confidence interval for the average monthly rate of change in fluoroscopy time at the Wilmington site is notably wide, indicating substantial uncertainty in the estimated trend. This high variability likely reflects limited or highly variable data for Wilmington, which reduces confidence in concluding whether fluoroscopy time is increasing or decreasing at this location. As a result, the trend estimates for Wilmington should be interpreted with caution, and further data collection or additional analysis may be necessary to clarify the true temporal pattern at this site.



Using the fitted Generalized Additive Model (GAM), fluoroscopy time trends for each clinical site six months beyond the observed data. The model provides predicted fluoroscopy times along with 95% confidence intervals to reflect uncertainty in these forecasts. These predictions, visualized as dashed lines with shaded confidence bands, extend the temporal patterns captured by the model and allow us to anticipate potential changes in fluoroscopy time over the near future. Such forecasting can guide ongoing evaluation of site performance and training effectiveness, while highlighting where further monitoring or intervention might be needed.